

Le rayonnement électromagnétique et son étude

M1 Astrophysique - CCP HAP702P Astrophysique

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Outline

1 The electromagnetic spectrum

2 Measurements in astrophysics

3 Bibliographic ressources

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1 The electromagnetic spectrum
- Information carriers in astrophysics
- Electromagnetic radiation

2 Measurements in astrophysics

3 Bibliographic ressources

Information carriers in astrophysics

- Electromagnetic radiation
  - Visible (optical) until 20th century
  - Full spectrum now
  - Main information carrier



M104 at visible-nIR wavelengths

Credit: ESO/P. Barthel

Information carriers in astrophysics

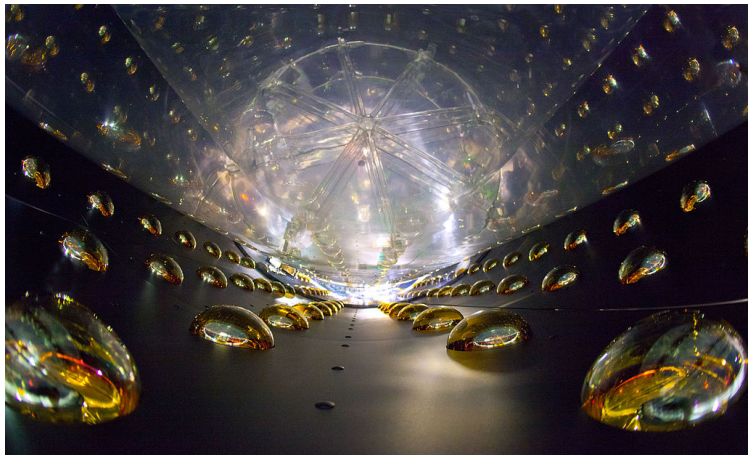
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  - Visible (optical) until 20th century
  - Full spectrum now
  - Main information carrier
- Cosmic rays
  - Coming from above atmosphere: Hess 1912
  - Origin: SNs and AGNs?



Credit: H.E.S.S. Collaboration, Frikkie van Greunen

Information carriers in astrophysics

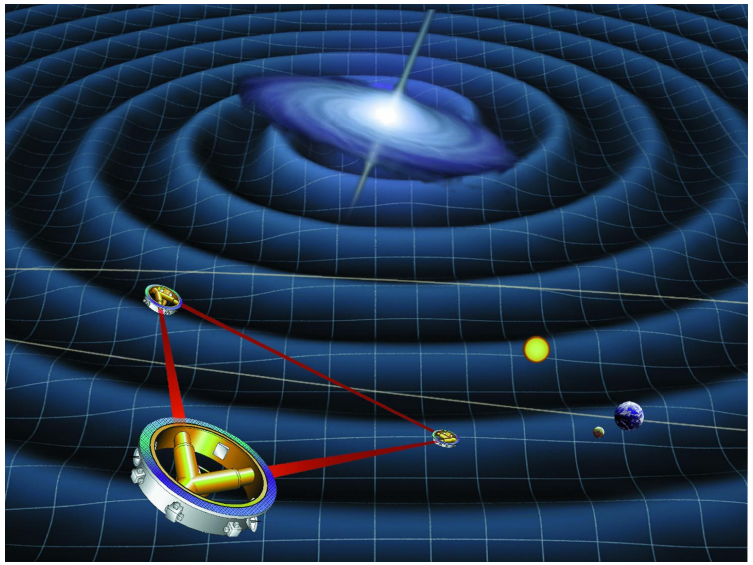
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  - Origin: SNs and AGNs?
- Neutrinos
  - Predicted by Fermi 1930, confirmed by Reines & Cowan 1956
  - Origin: Sun and SNs (cf. SN 1987A)



Credit: The Daya Bay Antineutrino Detector

Information carriers in astrophysics

- Electromagnetic radiation
  - Visible (optical) until 20th century
  - Full spectrum now
  - Main information carrier
- Cosmic rays
  - Coming from above atmosphere: Hess 1912
  - Origin: SNs and AGNs?
- Neutrinos
  - Predicted by Fermi 1930, confirmed by Reines & Cowan 1956
  - Origin: Sun and SNs (cf. SN 1987A)
- Gravitational waves
  - Predicted in 1916 by Einstein
  - Observed on 14th September 2015
  - Sources: binary compact objects, SNs

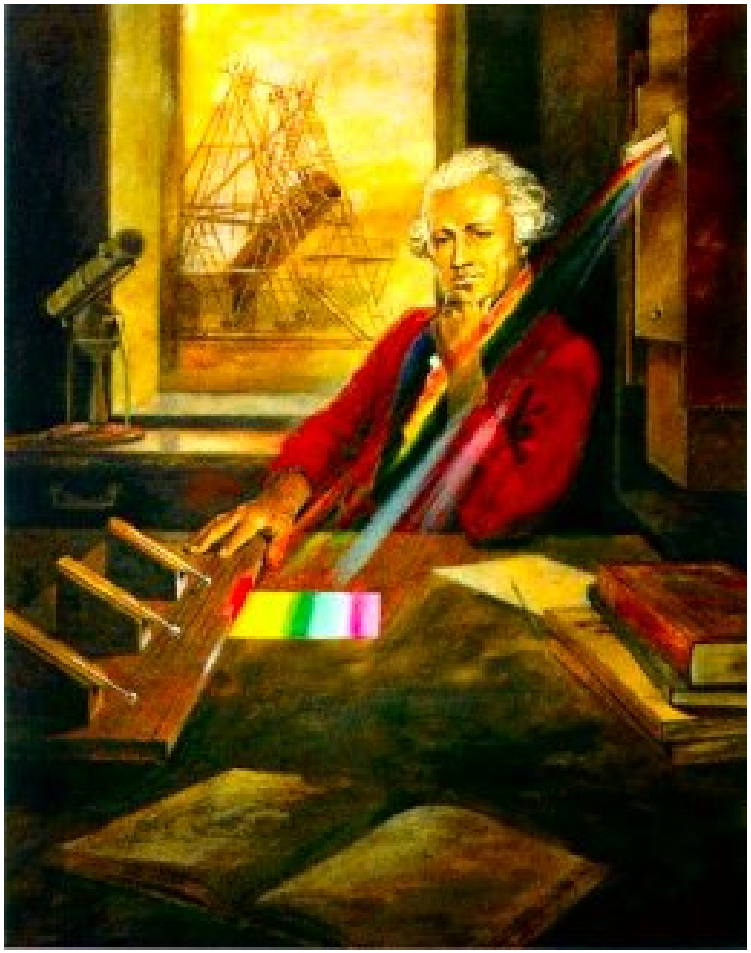


Artist's conception of LISA spacecraft

Credit: ESA/NASA

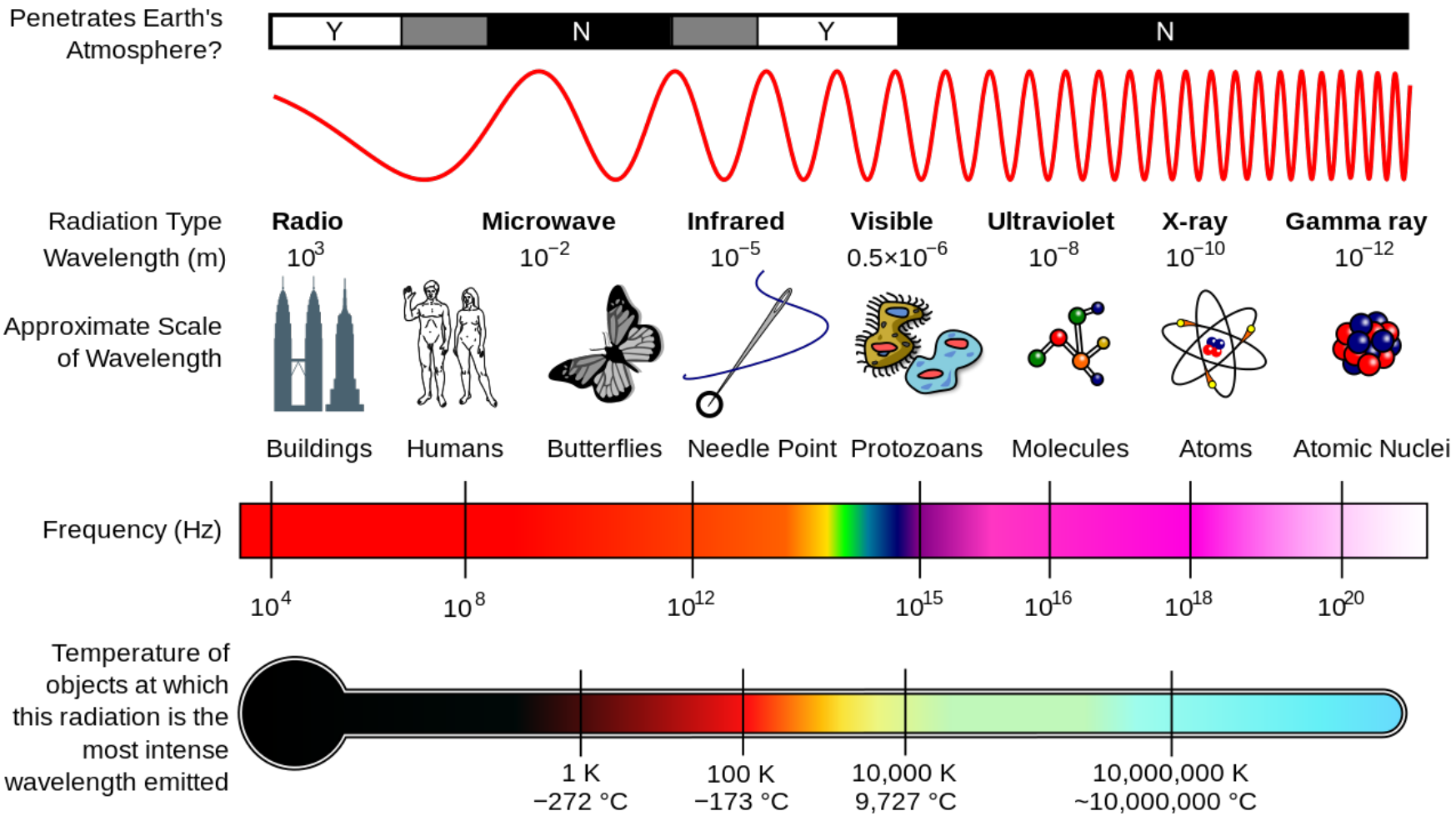
The electromagnetic spectrum (1/2)

- EM waves characterized by
  - Frequency: ν
  - Wavelength: $\lambda = \dfrac{c}{\nu}$
  - Energy per photon: $E = h\nu = \dfrac{hc}{\lambda}$
- Constants
  - $c = 299\,792\,458\ \mathrm{m}\cdot\mathrm{s}^{-1}$
  - $h = 6.626\times10^{-34}\mathrm{J}\cdot\mathrm{s}$



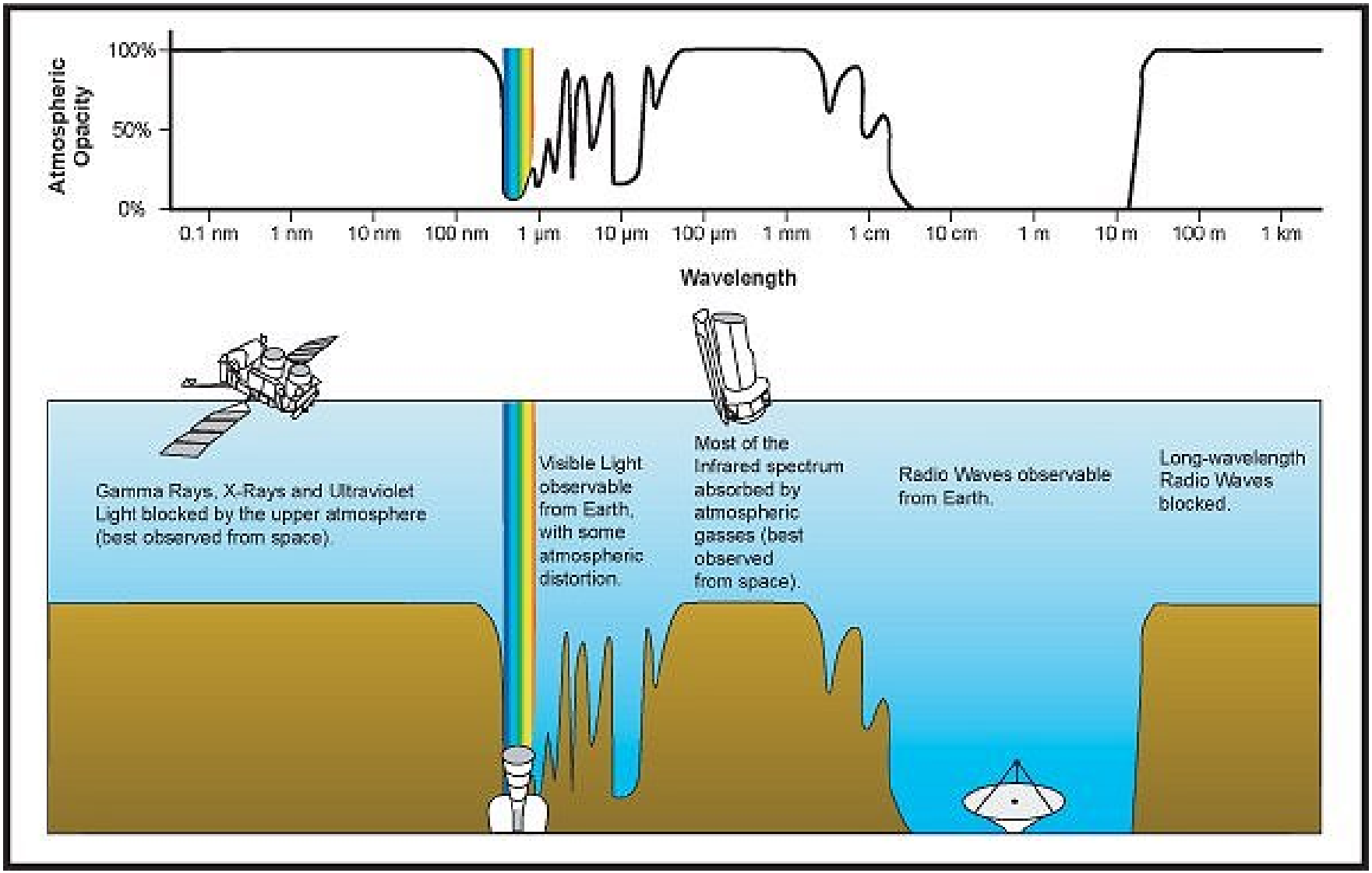
Herschel's experiment

The electromagnetic spectrum (2/2)



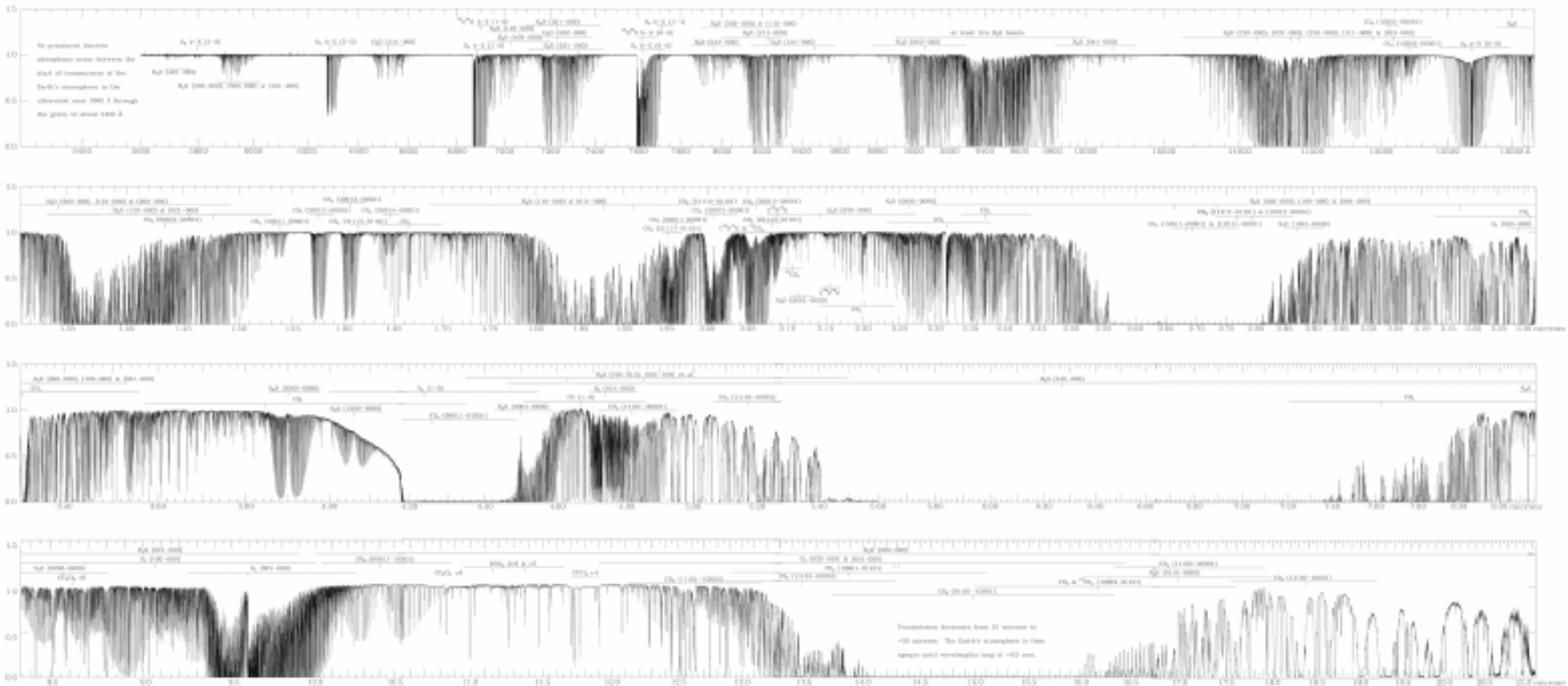
Credit: NASA

Atmospheric transmission (1/2)



Credit: Caltech

Atmospheric transmission (2/2)



Atmospheric transmission from 500 nm to 30 μm (Pen 2014, LRSP, 11, 2), data orig. from Hinkle et al. 2003, BAAS 35, 1260

Outline

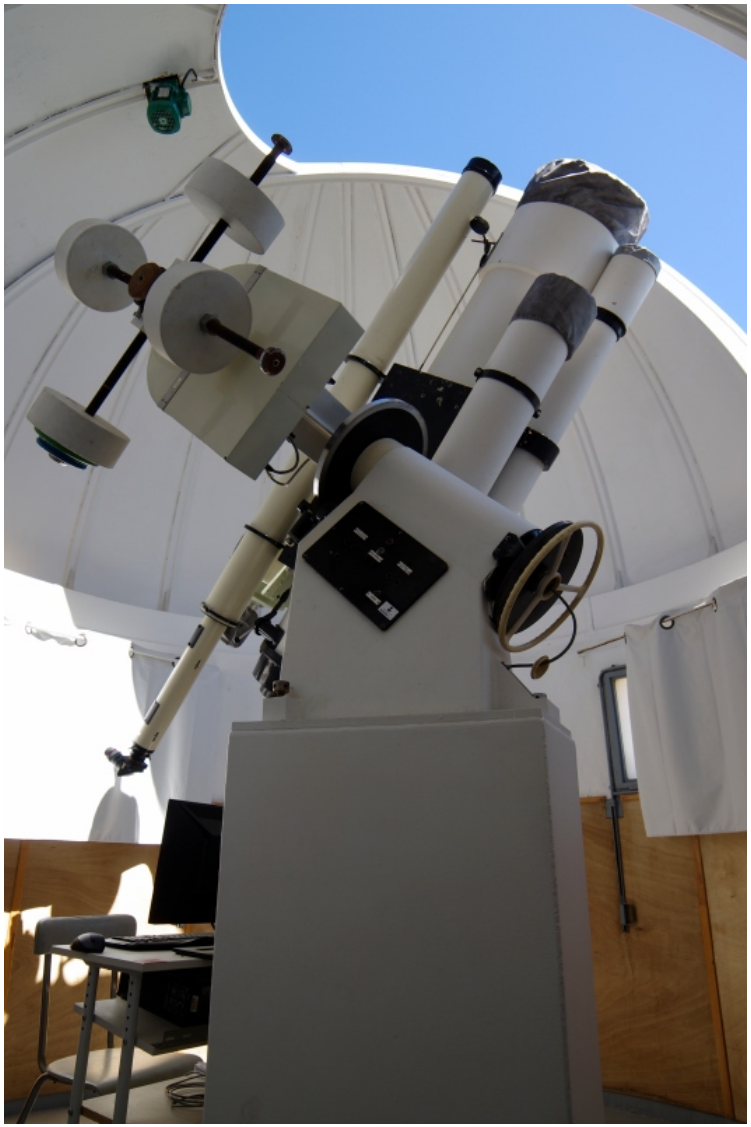
1 The electromagnetic spectrum

2 Measurements in astrophysics
- Telescopes
- Diffraction : image formation
- Diffraction limit: Rayleigh criterion
- Types of measurement

3 Bibliographic ressources

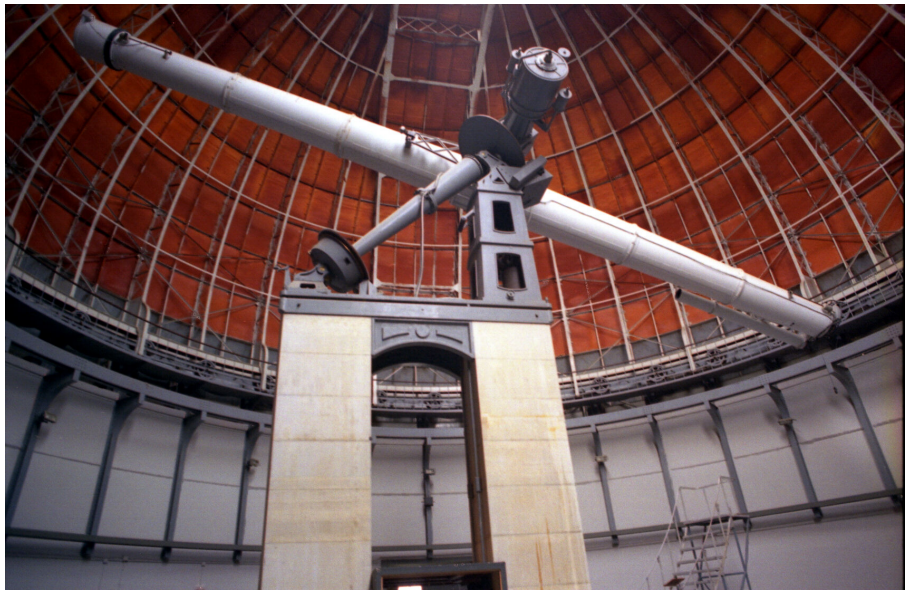
Basic principles of a telescope

- Collect luminous energy (photons)
  - Flux $\propto D^2$
- Resolve fine details (angle α)
  - Diffraction limit: $\alpha \propto 1/D$
  - Sampling (pixel size) / plate scale $\alpha \propto 1/f$



Optical design

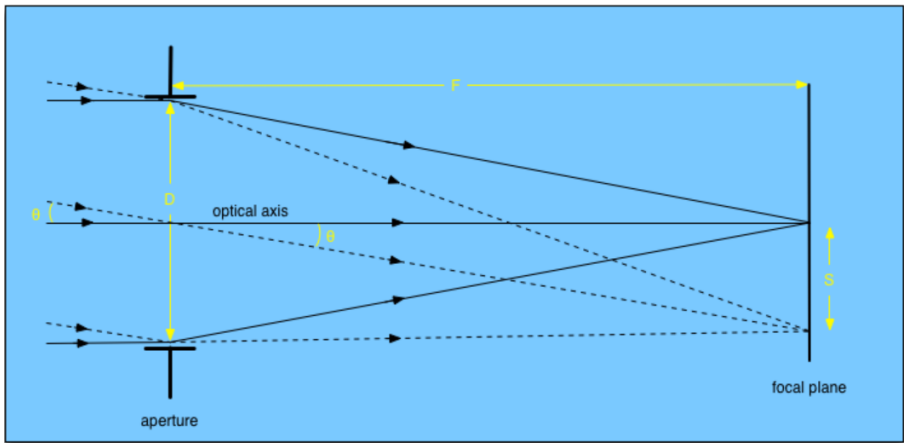
- Refractor



Large refractor in Nice

Optical design

- Refractor
  - converging dioptric objective
  - achromatic doublet, triplet

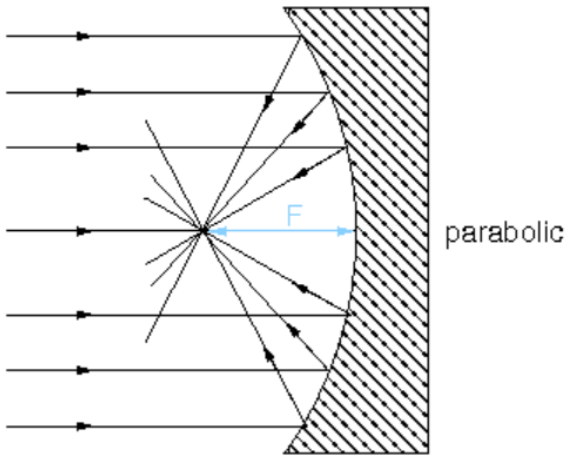


Schematic of a refractor

Credit: S. Littlefair

Optical design

- Refractor
  - converging dioptric objective
  - achromatic doublet, triplet
- Reflector
  - primary mirror: paraboloid



Schematic of a reflector's primary miror

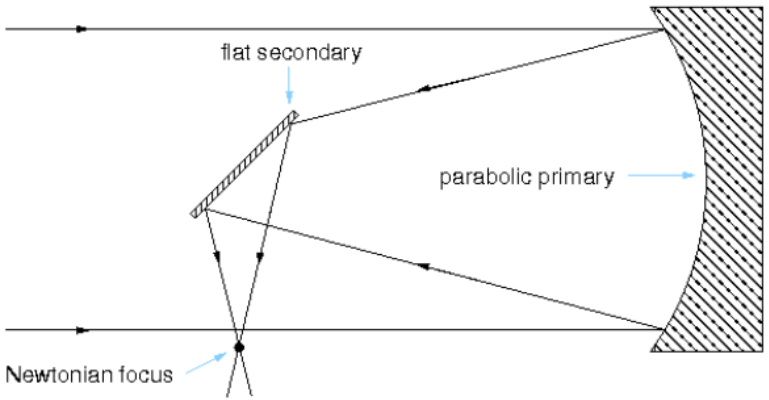
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- Reflector
  - primary mirror: paraboloid

➥ equivalent to refractor objective

  - generally also secondary mirror



Schematic of a Newton reflector

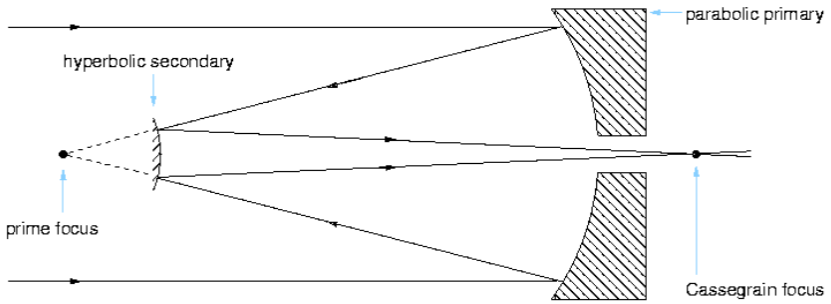
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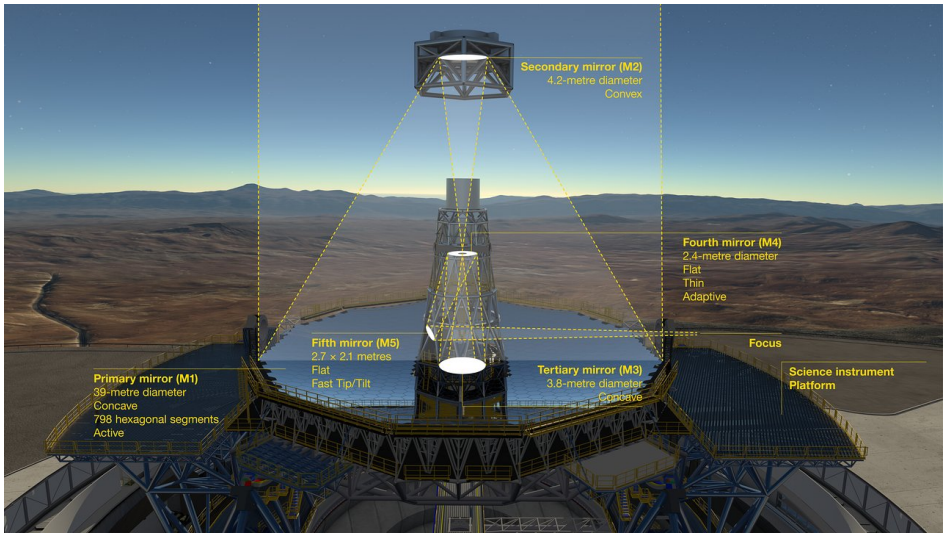


Schematic of a Cassegrain reflector

Credit: S. Littlefair

Optical design

- Refractor
  - converging dioptric objective
  - achromatic doublet, triplet
- Reflector
  - primary mirror: paraboloid
  - ➥ equivalent to refractor objective
  - generally also secondary mirror
- Aberrations
  - chromatism, sphericity, coma, astigmatism, distortions

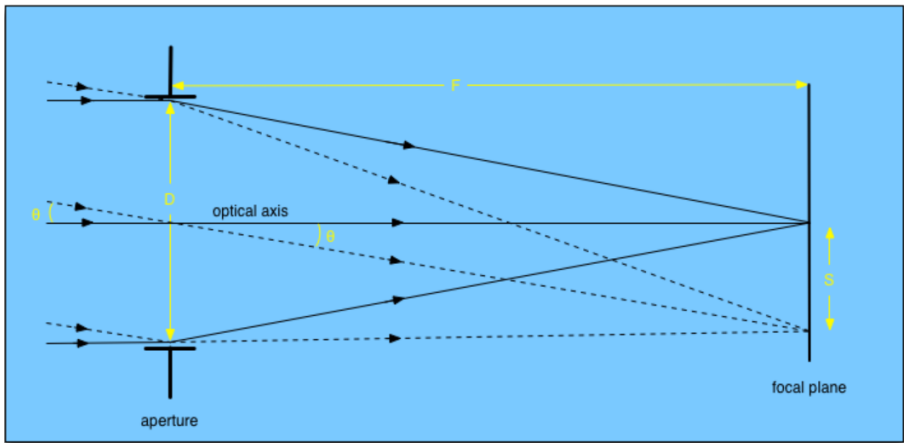


5-mirror optical system of ESO's Extremely Large Telescope (ELT)

Credit: ESO

Field of view and plate scale

- Refractor schematic; size in focal plane
  - $s = F \tan\theta \simeq F\theta$
- Field of view
  - viewing angle $\leftrightarrow$ detector size
- Platescale
  - viewing angle $\leftrightarrow$ pixel size
  - scale $p = \theta/s = 1/F$ (rad/m)
    $= 0.206265/F$ ("/μm, if F in m)



Schematic of a refractor

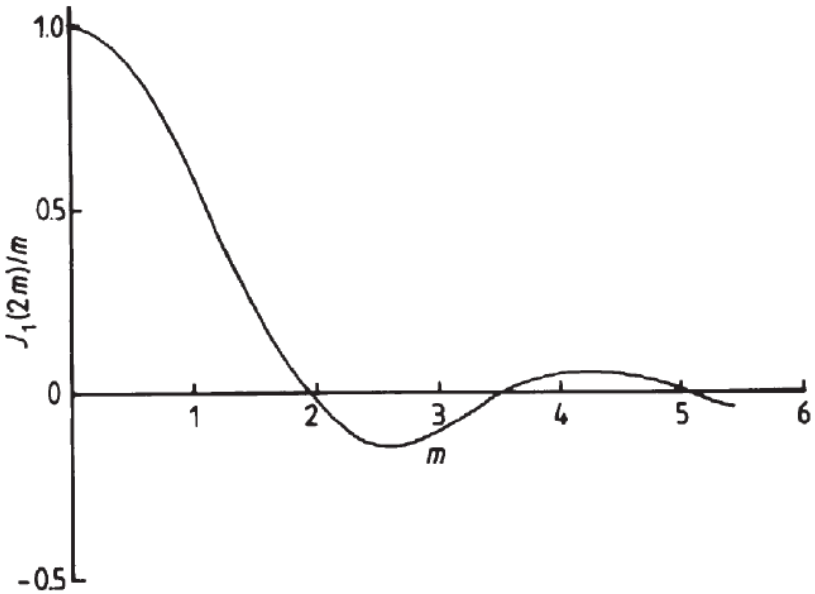
Credit: S. Littlefair

Diffraction limit

- Diffraction at infinity (Fraunhofer) by a circular aperture

$$I(\theta) = I_0 \left(\frac{2 J_1 \left(\frac{\pi D \theta}{\lambda} \right)}{\frac{\pi D \theta}{\lambda}} \right)^2$$

- J_1 : order 1 Bessel function
- 1st zero: $\theta = 1.22\,\lambda/D$

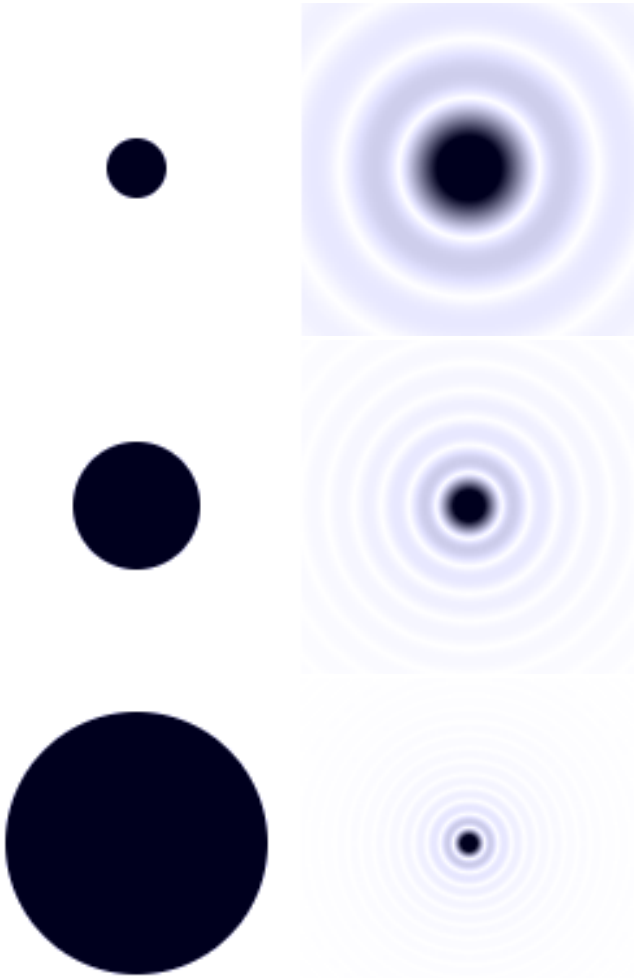


Bessel function J_1

Credit : Astrophysical Techniques, R. Kitchin

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  - J_1 : order 1 Bessel function
  - 1st zero: $\theta = 1.22\,\lambda/D$
- « Airy disc» : image of a point source at infinity by a circular aperture
- Also known as:
  - Point spread function (PSF)
  - Impulse response



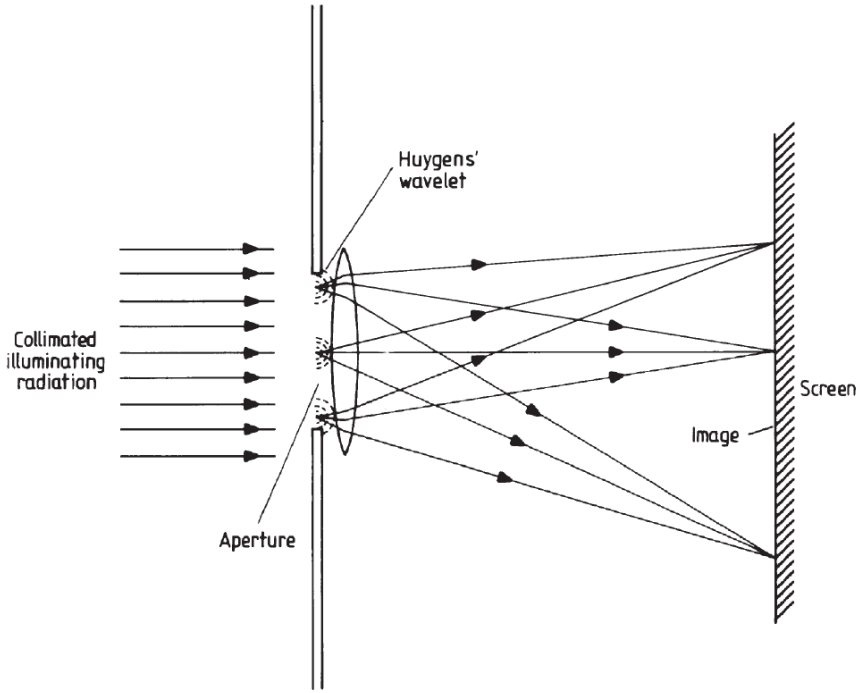
Apertures (left) and corresponding PSF (right)

Credit: Fenêtres sur l'univers

Diffraction : image formation

- Diffraction by the circular aperture of a telescope of diameter D
  - diameter of the central lobe of the diffraction spot in the image focal plane (NB: 86% of energy): $d = 2 \times 1.22\,\lambda F/D$
  - AN : telelens : $D = 50$ mm ; $F = 500$ mm ; $\lambda = 500$ nm

➔ $d = 12.2\ \mu$m ($\gtrsim$ pixel size)



Diffraction by the entrance pupil of a refractor

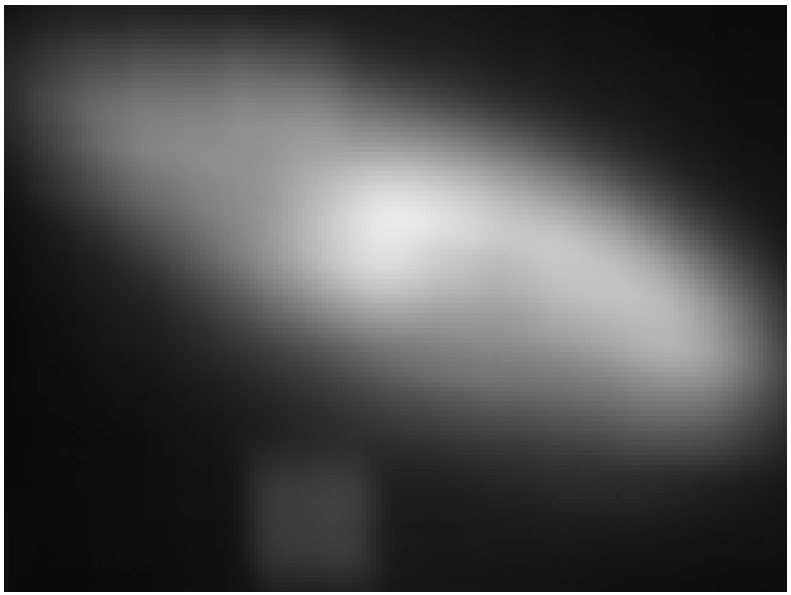
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- Image formed by an instrument
  1. image computed with geometric optics
  2. each point of the image is substituted with the PSF
     - Mathematically : convolution of the geometric image by the instrument's PSF



Simulated images of M31 with increasing D.

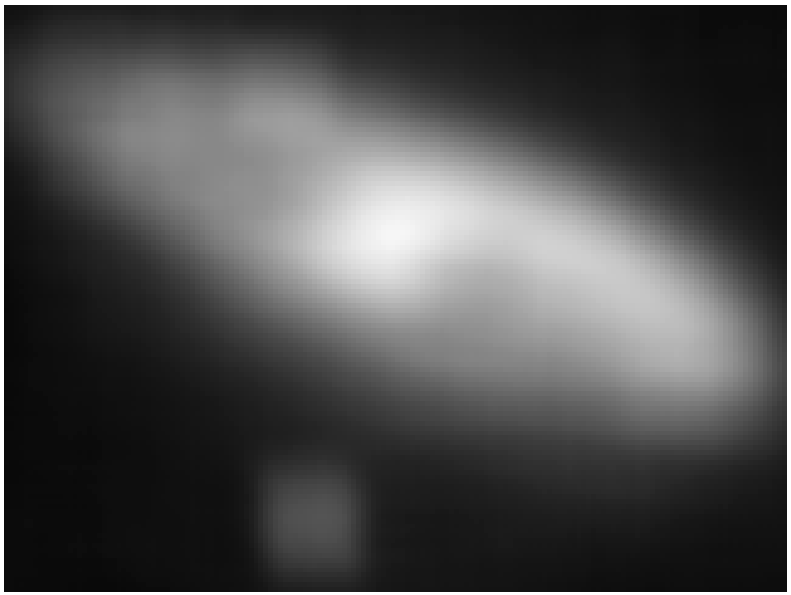
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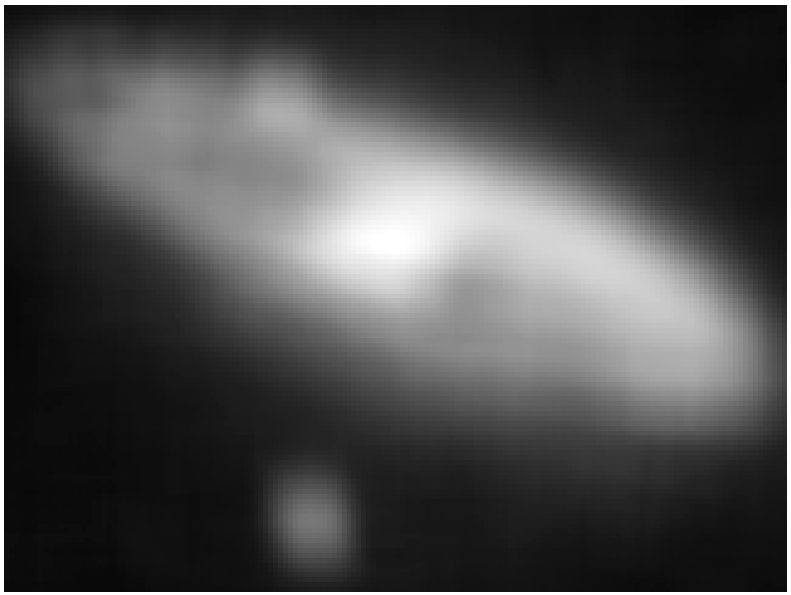
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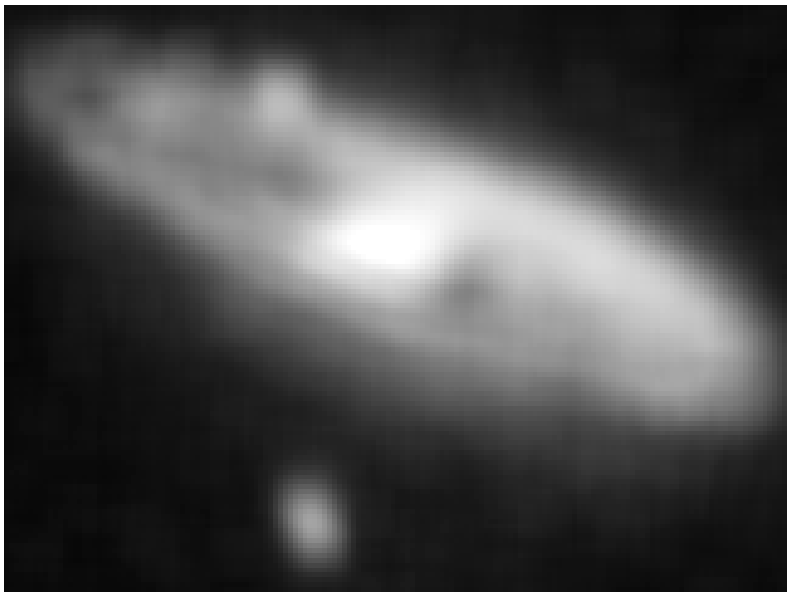
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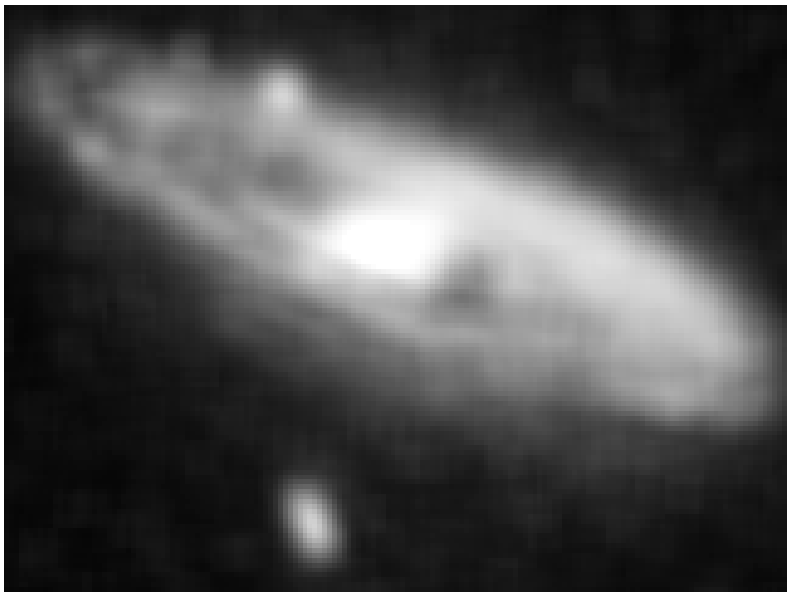
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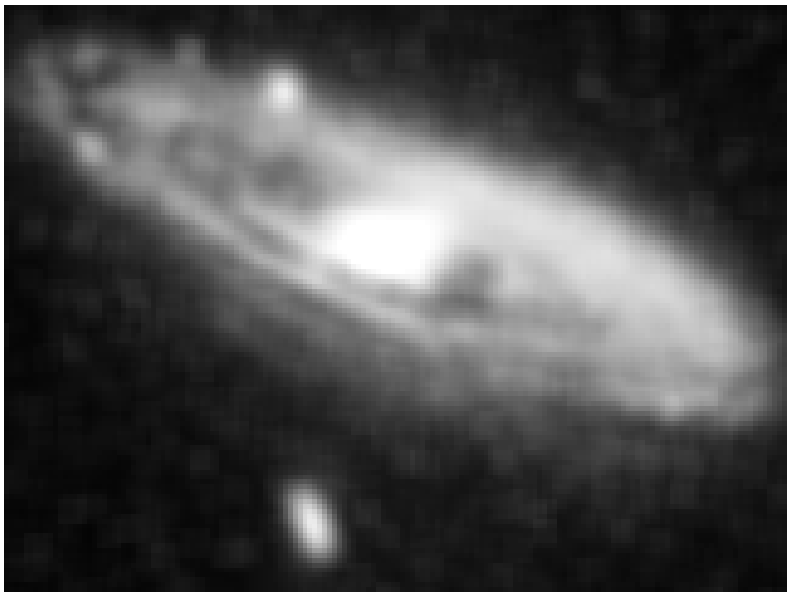
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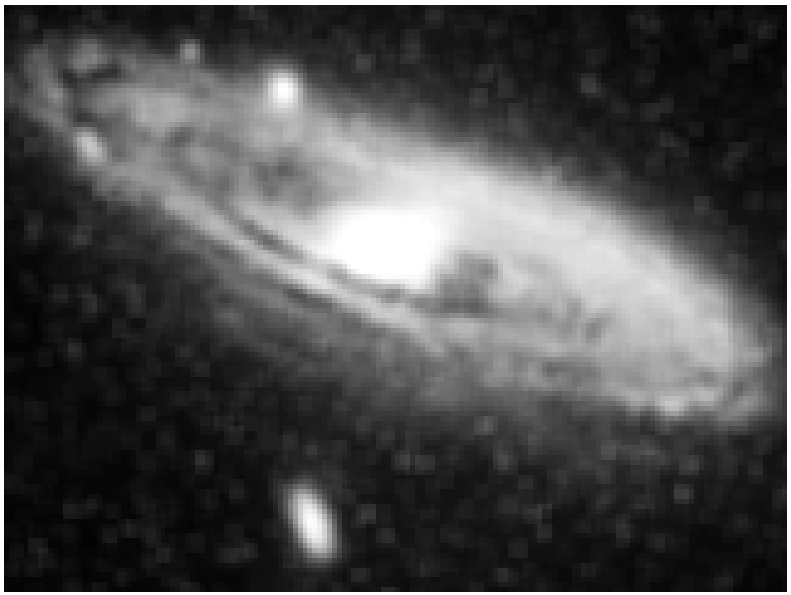
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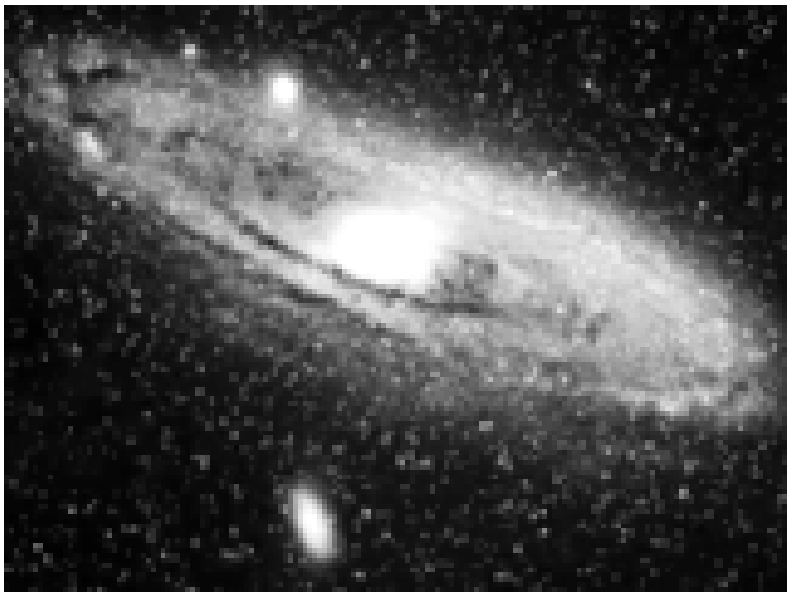
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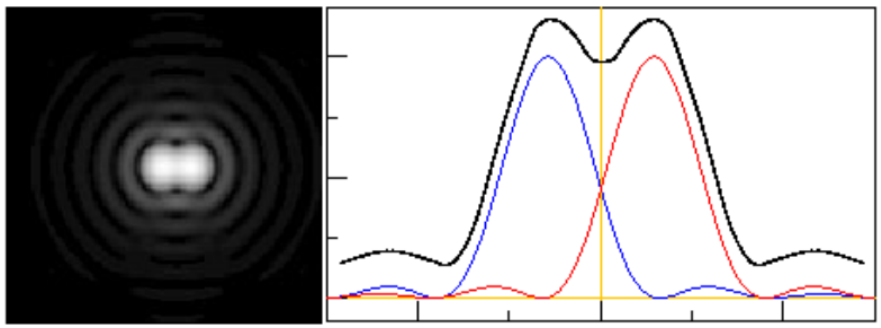
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Diffraction limit: Rayleigh criterion

- Resolving power of an instrument:

  minimal angular separation between two distinguishable point sources

- Rayleigh criterion
  - $\Delta\theta = 1.22\,\lambda/D$
  - Angle between sources = radius of central lobe of the PSF
  - Empirical criterion (beatable under specific circumstances)
- Valid for a perfect instrument



Appearance of sources at the Rayleigh criterion limit

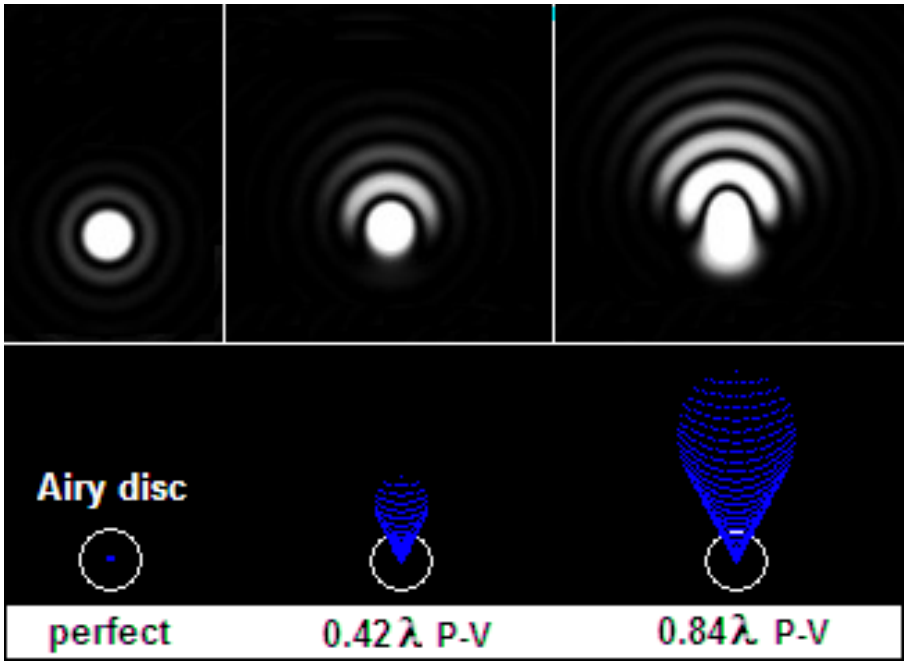
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  - Empirical criterion (beatable under specific circumstances)
- Valid for a perfect instrument
  - Imperfections of optical formula
  - Polishing precision of optical parts



Coma aberration
Top : PSF ; Bottom : raytracing ; increasing aberration rightwards

Credit: Amateur Telescope Optics

Diffraction limit: Rayleigh criterion

- Resolving power of an instrument:

  minimal angular separation between two distinguishable point sources

- Rayleigh criterion
  - $\Delta\theta = 1.22\,\lambda/D$
  - Angle between sources = radius of central lobe of the PSF
  - Empirical criterion (beatable under specific circumstances)
- Valid for a perfect instrument
  - Imperfections of optical formula
  - Polishing precision of optical parts
  - Presence of secondary miror (télescope)
  - Atmospheric turbulence



Proxima Cen as pictured by HST
The diffraction spikes due to the secondary miror support are clearly visible.

Credit : NASA/ESA

Effects of the atmosphere

- Extinction
  - specially at X, UV, IR wavelengths.
  - effects of clouds, particles
  - airmass: $am \approx sec(z) = 1/\cos(z)$, with z the zenith angle
- Scattering (Rayleigh, Mie)
- Emission
  - thermal emission IR
  - airglow (e.g. N+O$\rightarrow$NO, OH, Na)

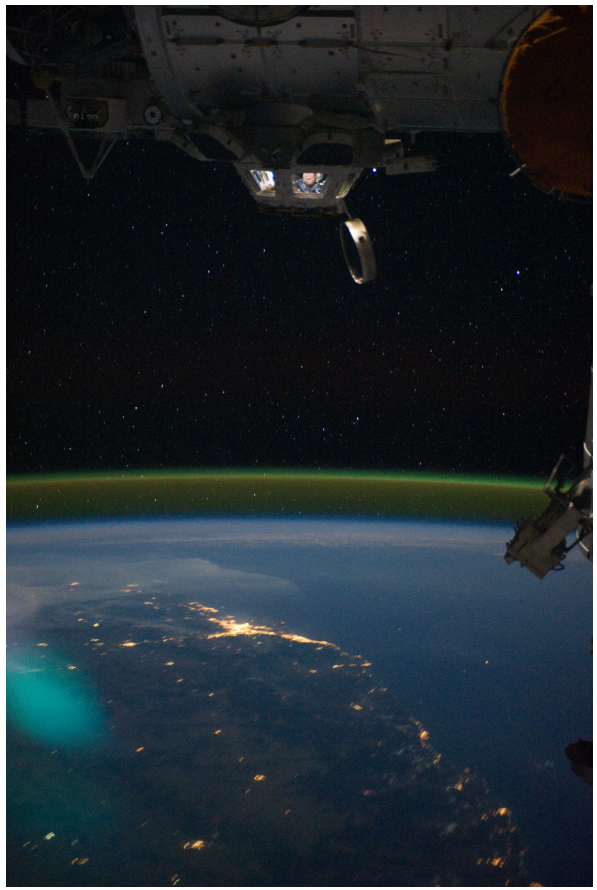


Airglow at VLT

Credit: Y. Beletsky (LCO)/ESO

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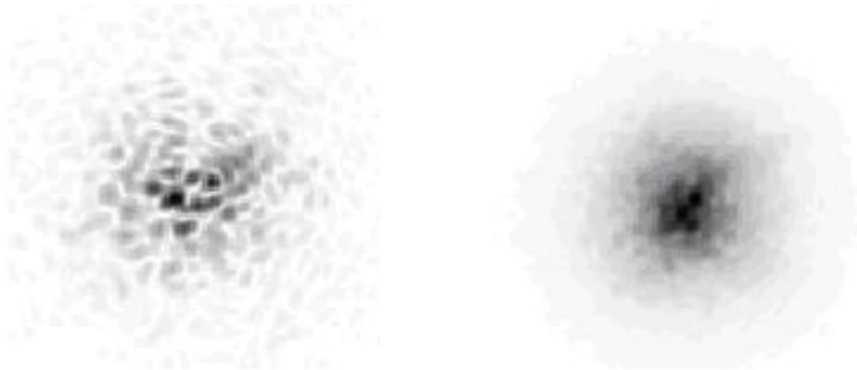


Airglow seen from ISS

Credit: NASA

Effects of the atmosphere

- Extinction
  - specially at X, UV, IR wavelengths.
  - effects of clouds, particles
  - airmass: $am \approx sec(z) = 1/\cos(z)$, with z the zenith angle
- Scattering (Rayleigh, Mie)
- Emission
  - thermal emission IR
  - airglow (e.g. N+O$\rightarrow$NO, OH, Na)
- Turbulence
  - flickering
  - image blurring

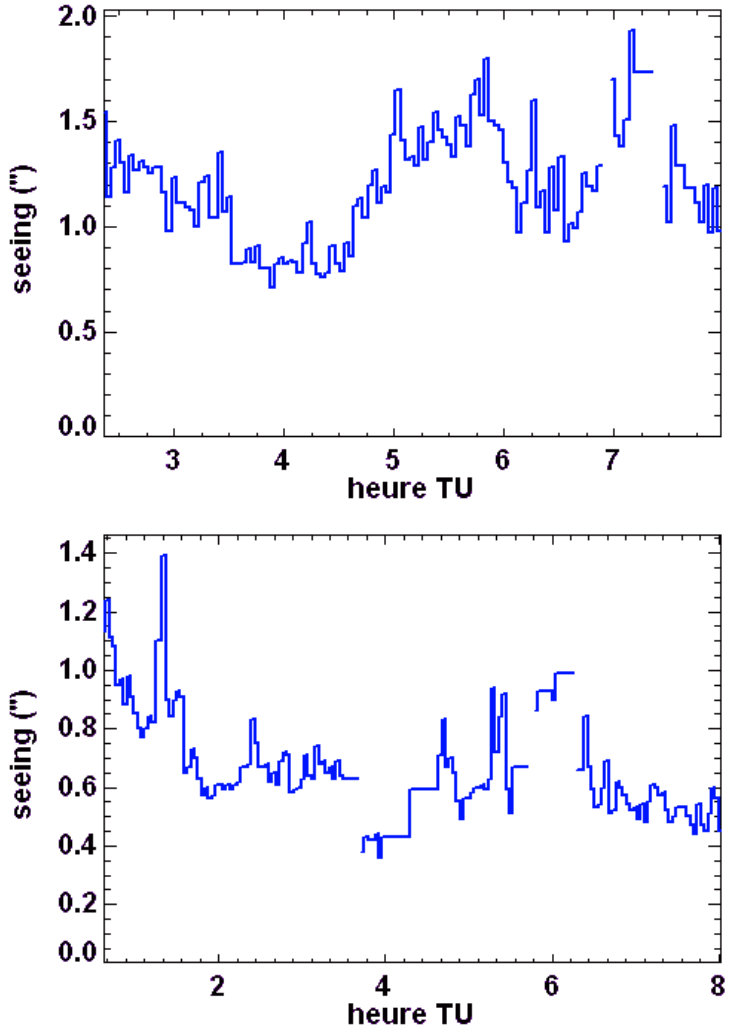


Speckles (short integration) and corresponding long integration stellar image

Credit: ESO

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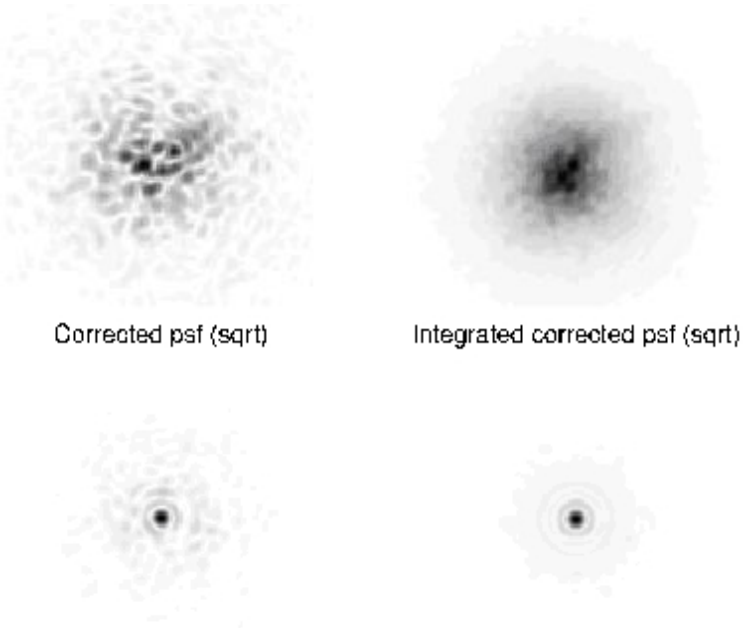
Seeing evolution for two different nights

Credit: ESO

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➥ adaptive optics (AO)



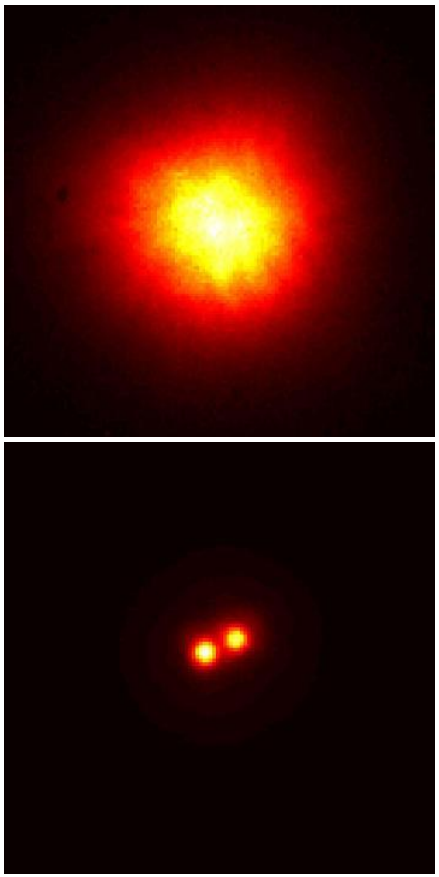
AO-corrected PSF

Credit: ESO

Effects of the atmosphere

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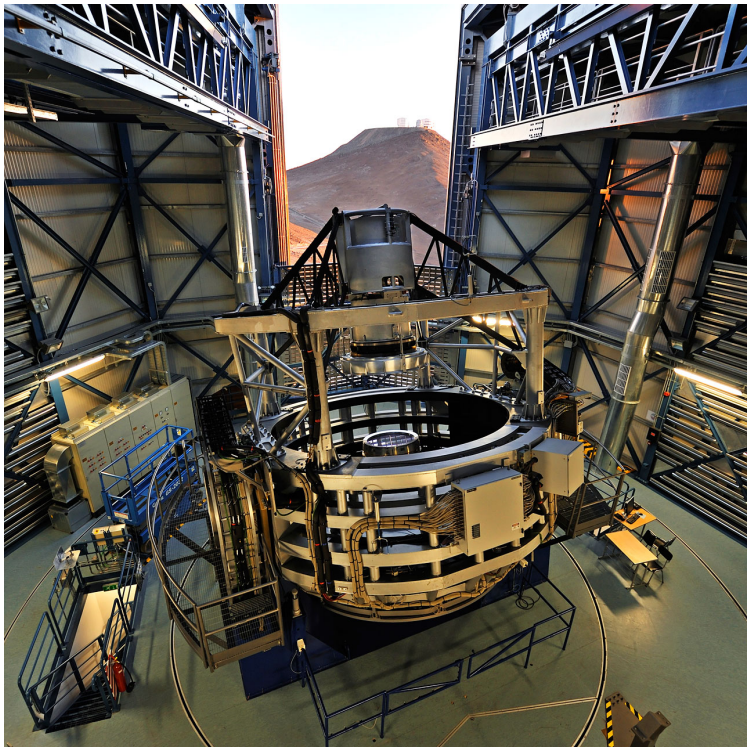
AO-corrected binary (bottom)

Credit: ESO

Types of measurement (1/2)

- Measurable photon properties

1. Incidence direction
2. Arrival time
3. Energy
4. Polarization
5. Bunching properties (e.g. intensity interferometer)



Visible and Infrared Survey Telescope for Astronomy

Credit: G. Hüdepohl/ESO

Types of measurement (2/2)

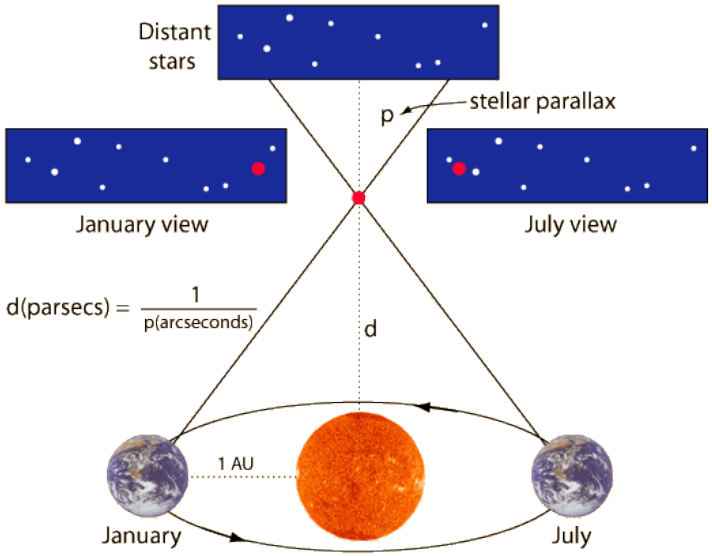
- Types of measurements
  - Astrometry: positions and motions
  - Photometry: fluxes in a filter system
  - Imaging: mapping, morphology
  - Spectroscopy: spectral energy distribution
  - Polarimetry: polarization degree, orientation, handedness
  - Combination
    - spectrophotometry, spectropolarimetry
  - Specific techniques
    - coronagraphy, interferometry



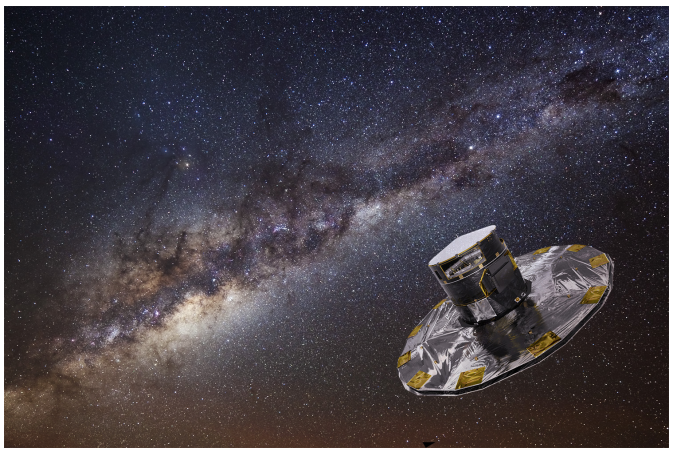
VISTA's camera
(0.9 - 1.2 μm 67 10^6 pixels)
Credit: ESO

Types of measurement (2/2)

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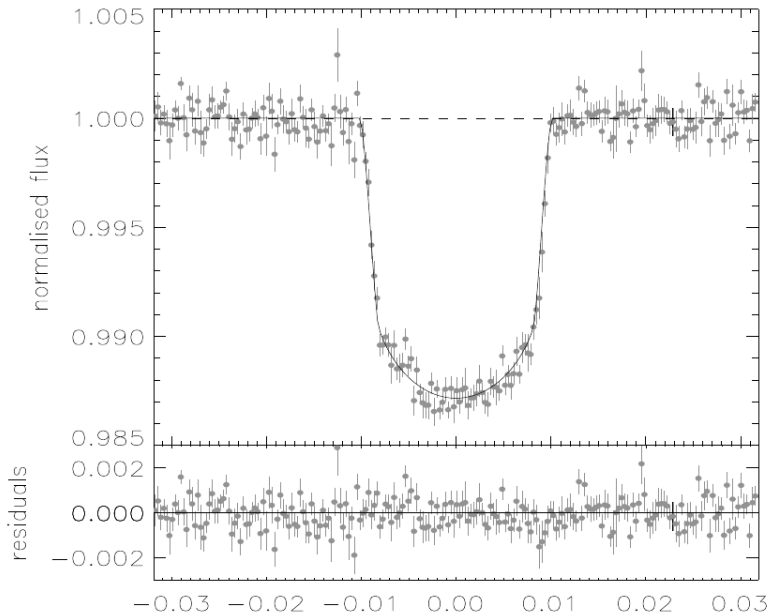
Annual parallax – Credit: Hyperphysics



Gaia – Credit: ESA

Types of measurement (2/2)

- Types of measurements
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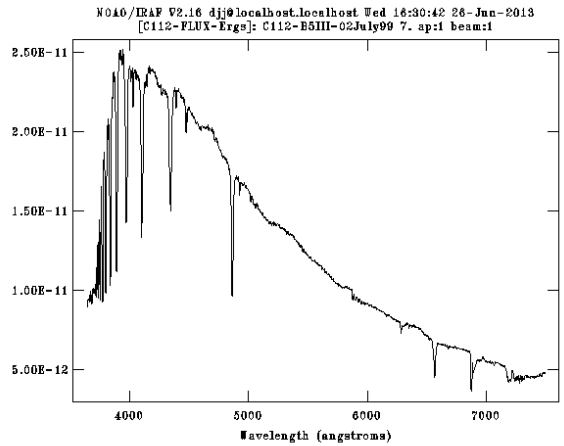


Transit of Corot-4b

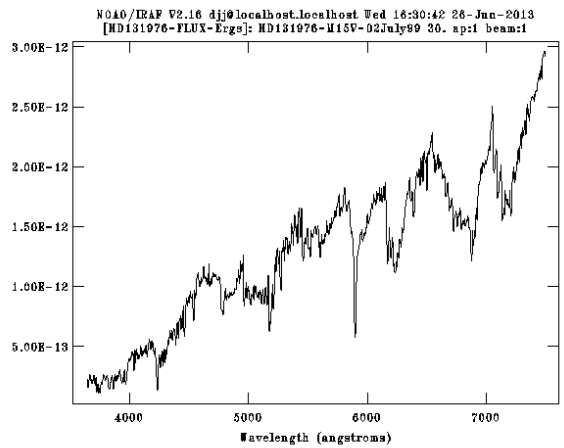
Aigrain et al. (2008)

Types of measurement (2/2)

- Types of measurements
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B5III star

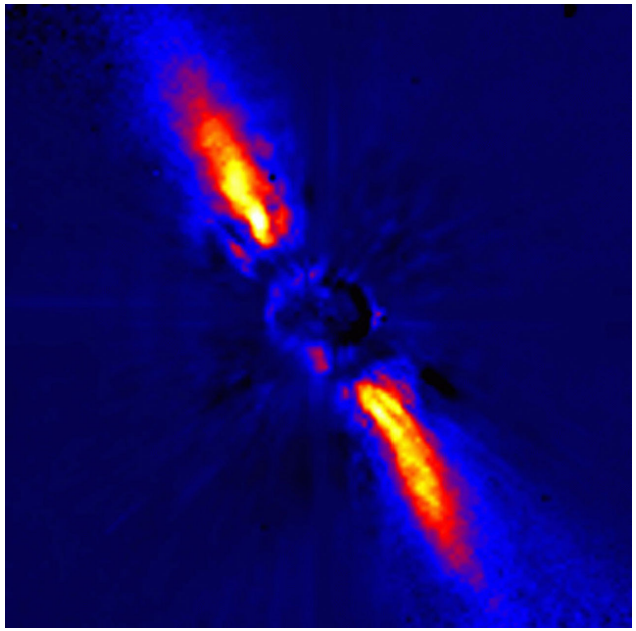


M1.5V star

James et al. (2013)

Types of measurement (2/2)

- Types of measurements
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Circumstellar disc of β Pictoris

Credit: ESO

Outline

1 The electromagnetic spectrum

2 Measurements in astrophysics

3 Bibliographic ressources

Bibliographic ressources

- *"Astronomie Astrophysique, Introduction, 5e édition"*, A. Acker, 2013, Dunod
  ISBN-13: 978-2100576401
  - Chapitre 2 : "Déchiffrer le rayonnement des astres"

- *"L'Observation en Astrophysique"*, P. Léna et al., 2008, EDP Sciences
  ISBN-13: 978-2868838773
- *"Astrophysical Techniques"*, C.R. Kitchin, 2013, CRC Press
  ISBN-13: 978-1466511156